An Extra Boost to Boosted Models

Matthew Yang

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1 Introduction

Gradient boosted models are built from a sequence of decision trees, where each tree is fitted to the residual errors of the ensemble constructed so far rather than to the original targets [1]. This stage-wise additive formulation, together with highly engineered implementations such as XGBoost [2], LightGBM [3] and CatBoost [4], has made boosting the default choice for tabular prediction tasks, where it remains competitive with, and frequently superior to, deep learning [5, 6].

Boosted models are, however, computationally expensive to train. Inducing a single tree requires evaluating candidate split points over the training set at every node, so the cost of each boosting iteration scales with both the number of instances and the number of features. Approximate split finding, histogram binning, gradient-based sampling and feature bundling all reduce this cost substantially [2, 3, 7], but they lower the constant factor rather than remove the underlying dependence on the full training set.

Training is also largely sequential. The sequential dependency is intrinsic to boosting itself, since the targets of iteration t are defined by the predictions of iterations $1, \ldots, t-1$; it is compounded by the greedy, top-down induction of each individual tree, in which a node cannot be split until its parent has been. Existing implementations therefore parallelise *within* a boosting iteration, over features or histogram bins, rather than across iterations. This stands in contrast to ensembles such as random forests [8], whose members are independent and thus embarrassingly parallel, but which forgo the error-correcting behaviour that gives boosting its accuracy.

With the ever-growing need to train models on larger datasets, more computationally efficient alternatives to tree-based boosting are needed.

2 Background and Related Work

Boosting originates in the observation that a set of weak learners can be combined into an arbitrarily strong one, formalised by AdaBoost [9] and later generalised to arbitrary differentiable losses as gradient boosting [1]. Decision trees are the conventional weak learner, but nothing in the gradient boosting framework requires them: any learner capable of fitting the negative gradient of the loss can be used.

This observation is particularly relevant to time series classification, where random-feature methods have shown that the expensive, data-dependent search performed by tree induction can often be replaced by cheap, data-independent random transformations. ROCKET [10] applies a large bank of random convolutional kernels followed by a linear classifier, achieving accuracy comparable to far more expensive ensembles such as HIVE-COTE [11] at a small fraction of the training cost. MiniRocket [12] and Hydra [13] refine this approach further. The underlying principle of approximating a costly learned representation with randomised features is the same one that motivates random feature approximations to kernel machines [14], and it is the principle we intend to carry into the boosting setting.

3 Aims and Research Questions

This research aims to investigate how parallelism can be introduced into the training of boosted models: first, by partitioning the data without inducing an explicit decision tree, and second, by training multiple boosted models concurrently and averaging their predictions, as in bootstrap aggregating (bagging) [15].

Specifically, we ask:

1. Can the decision tree conventionally used as the weak learner in gradient boosting be replaced by a parallelisable partitioning algorithm?
2. What is the resulting effect on training time, and how does this approach scale with the number of training instances?
3. What accuracy, if any, is sacrificed relative to state-of-the-art boosted models on standard benchmarks?

4 Methodology

This section briefly outlines how we will investigate our proposed research.

4.1 Replication

We first attempt to replicate the structure of current state-of-the-art models, in order to establish a baseline against which our own results can be compared.

4.2 Implementation

We will mimic the structure of current state-of-the-art models, replacing the tree-based weak learner with a parallel partitioning algorithm and making further changes where necessary.

4.3 Benchmarking

The UCR Time Series Classification Archive [16] will provide a quantitative comparison of our models' performance against existing methods, using the archive's default train/test splits so that our results are directly comparable to published ones [17]. We will report both classification accuracy and training time. Accuracy across multiple datasets will be compared using the Friedman test with post-hoc pairwise comparisons, as recommended by Demšar [18], and summarised with critical difference diagrams.

5 Timeline

Week:	1	2	3	4	5	6	7	8	9	10	11	12
Initial meeting		•										
Review of literature	•	•	•									
Writing proposal		•	•									
Replication of current research				•								
Implementation of proposal					•	•	•	•	•	•		
Benchmarking models						•	•	•	•	•	•	
Project presentation												•
Final report			•	•	•	•	•	•	•	•	•	•

Table 1: Project schedule by week.

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